$$B \quad = \frac{5x+12}{x+3}\left[x=-2\right] = \frac{2}{1} = 2$$

$$\frac{5x+12}{x^2+5x+6} = \left[\frac{3}{x+3}+\frac{2}{x+2}\right]$$

$$\Delta\left[\frac{5x+12}{x^2+5x+6}\right] = \Delta\left[\frac{3}{x+3}+\frac{2}{x+2}\right]$$

$$= \left[\frac{3}{x+1+3}-\frac{3}{x+3}\right]+\left\{\frac{2}{x+1+2}-\frac{2}{x+2}\right\}$$

$$= 3\left[\frac{1}{x+4}-\frac{1}{x+3}\right]+2\left[\frac{1}{x+3}-\frac{1}{x+2}\right]$$

$$= \left[\frac{-3}{(x+4)(x+3)}-\frac{2}{(x+3)(x+2)}\right]$$

$$= \frac{-5x-14}{(x+2)(x+3)(x+4)}$$

Example 5.6

Evaluate $\Delta^2\left(\frac{1}{x}\right)$ by taking '1' as the interval of differencing.

Solution:

$$\Delta^2\left(\frac{1}{x}\right) = \Delta\left(\Delta\left(\frac{1}{x}\right)\right)$$

Now $\Delta\left[\frac{1}{x}\right] = \frac{1}{x+1}-\frac{1}{x}$

$$\Delta^2\left(\frac{1}{x}\right) = \Delta\left(\frac{1}{1+x}-\frac{1}{x}\right)$$

$$= \Delta\left(\frac{1}{1+x}\right)-\Delta\left(\frac{1}{x}\right)$$

Similarly $\Delta^2\left(\frac{1}{x}\right) = \frac{2}{x(x+1)(x+2)}$

DO YOU KNOW?

$$\Delta^n\left(\frac{1}{x}\right) = \frac{(-1)^n n!}{x(x+1)(x+2).....(x+n)}$$

Example 5.7

Prove that $f(4) = f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1)$ taking '1' as the interval of differencing.

Solution:

We know that $f(4) - f(3) = \Delta f(3)$

$$f(4) - f(3) = \Delta f(3)$$

$$= \Delta\left[f(2) + \Delta f(2)\right] \quad \because \left[f(3) - f(2) = \Delta f(2)\right]$$

$$= \Delta f(2) + \Delta^2 f(2)$$

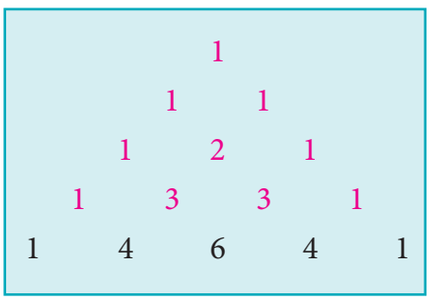
$$= \Delta f(2) + \Delta^2\left[f(1) + \Delta f(1)\right]$$

$$\therefore f(4) = f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1).$$

Example 5.8

Given $U_0 = 1, U_1, = 11, U_2 = 21, U_3 = 28$ and $U_4 = 29$ find $\Delta^4 U_0$

Solution :



$$\Delta^4 U_0 = (E-1)^4 U_0$$

$$= \left(E^4 - 4E^3 + 6E^2 - 4E + 1\right) U_0$$

$$= E^4U_0 - 4E^3U_0 + 6E^2U_0 - 4EU_0 + U_0$$

$$= U_4 - 4U_3 + 6U_2 - 4U_1 + U_0$$

$$= 29 - 4(28) + 6(21) - 4(11) + 1.$$

$$= 156 - 156 = 0$$

Example 5.9

Given $y_3 = 2,\ y_4 = -6,\ y_5 = 8, y_6 = 9$ and $y_7 = 17$ Calculate $\Delta^4 y_3$

Solution :

Given $y_3 = 2,\ y_4 = -6,\ y_5 = 8, y_6 = 9$ and $y_7 = 17$

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